LINEAR ALGEBRA · LESSON 10

The Four Fundamental Subspaces

Every matrix carries four spaces: what you can **reach**, what does **nothing**, what you can **measure**, and the **hidden link** between your rules.

r, n − r, r, m − r | [A | I] → [R | E] | MIT 18.06, Lecture 10

THE FOUR

Two you know, two new

Space	Lives in	Meaning
$C(A)$ columns	$\mathbb{R}^m$	reachable targets
$N(A)$	$\mathbb{R}^n$	do-nothing trades
$C(A^T)$ rows	$\mathbb{R}^n$	what the rules can measure
$N(A^T)$ left null	$\mathbb{R}^m$	rule combos that are always 0

m = rules (rows), n = funds (columns).

DIMENSIONS

r, n – r, r, m – r

$$\dim C(A) = r \qquad \dim N(A) = n - r$$

$$\dim C(A^T) = r \qquad \dim N(A^T) = m - r$$

Row space and column space have the **same** dimension: each pivot sits in one row and one column.

ONE REDUCTION

[A | I] → [R | E]

- $C(A)$: pivot columns of A
- $N(A)$: special solutions from R
- $C(A^T)$: non-zero rows of R
- $N(A^T)$: rows of E next to R's zero rows

$EA = R$ records every row operation.

All four, worked

$$[R \mid E] = \left[\begin{array}{ccc|ccc} 1 & 0 & 0.5 & -1.4 & 0.2 & 0 \\ 0 & 1 & 0.5 & 2.4 & -0.2 & 0 \\ 0 & 0 & 0 & 1.4 & -0.2 & 1 \end{array}\right]$$

$C(A)$: Nifty, bonds. $N(A)$: $(-0.5, -0.5, 1)$. $C(A^T)$: rows of R. $N(A^T)$: $(1.4, -0.2, 1) \propto (7, -1, 5)$.

What the rules can measure

Rows of R: **Nifty content** $x_1 + 0.5x_3$ and **bond content** $x_2 + 0.5x_3$.

- budget = Nifty content + bond content
- return = 12 × Nifty content + 7 × bond content
- exposure = Nifty content

Three rules, only **two** independent measurements.

The hidden link

$y = (7, -1, 5)$ gives 0 for every fund:

Nifty $7 - 12 + 5 = 0$, bonds $7 - 7 + 0 = 0$, balanced $7 - 9.5 + 2.5 = 0$

SO FOR EVERY PORTFOLIO

$7B - R + 5E = 0$, i.e. $R = 7B + 5E$: the test for which targets are reachable.

THE PAIRS

n = r + (n – r), m = r + (m – r)

Trade side ($\mathbb{R}^n$): measured (row space) + invisible (null space).

Target side ($\mathbb{R}^m$): reachable (column space) + blocked by hidden links (left null space).

Four funds: $4 = 2 + 2$ and $3 = 2 + 1$.

IN MARKETS

A complete market

Futures + 20,000 call: left null $(0, 1, 0)$, "payoff at 20,000 is always 0". That scenario can't be touched.

Add the 19,900 call: $N(A^T) = \{0\}$. Every payoff reachable.

> COMPLETE MARKET
> Left null space = $\{0\}$: no hidden link, every target reachable.

RECAP

Key takeaways

1. Four spaces: $C(A)$, $N(A)$, $C(A^T)$, $N(A^T)$.
2. Dimensions r, $n - r$, r, $m - r$.
3. One reduction: $[A \mid I] \to [R \mid E]$.
4. Row space: rows of R. Column space: columns of A .
5. Left null space = the hidden link = the target test.

QUANT FROM FIRST PRINCIPLES

The full interactive lesson is free

A menu picker showing all four subspaces, a four-subspaces lab for any matrix, and worked hedging and options examples:

learn.krisarj.com

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